

Instructor(s): Korytov/Mueller

PHYSICS DEPARTMENT

PHY2049, Spring 2020

Final Exam

April 28, 2020

Name (print): _____

Signature: _____

*On my honor, I have neither given nor received unauthorized aid on this examination.***YOUR TEST NUMBER IS THE 5-DIGIT NUMBER AT THE TOP OF EACH PAGE.****DIRECTIONS**

- (1) **Code your test number on your answer sheet (use 76–80 for the 5-digit number).** Code your name on your answer sheet. **Darken circles completely (errors can occur if too light).** Code your student number on your answer sheet.
- (2) Print your name on this sheet and sign it also.
- (3) Do all scratch work anywhere on this exam that you like. At the end of the test, this exam printout is to be turned in. No credit will be given without both answer sheet and printout with scratch work.
- (4) Work the questions in any order. Incorrect answers are not taken into account in any way; you may guess at answers you don't know.
- (5) If you think that none of the answers is correct, please choose the answer given that is closest to your answer.
- (6) **Blacken the circle of your intended answer completely, using a number 2 pencil.** Do not make any stray marks or the answer sheet may not read properly. Completely erase all incorrect answers, or take a new answer sheet.
- (7) As an aid to the examiner (and yourself), in case of poorly marked answer sheets, please circle your selected answer on the examination sheet. Please remember, however, that in the case of a disagreement, the answers on the bubble sheet count, NOT what you circle here. Good luck!!!

>>>>>>>**WHEN YOU FINISH**<<<<<<<

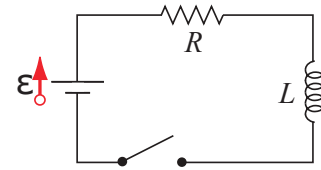
Hand in the answer sheet separately.

Constants		
$m_e = 9.11 \times 10^{-31} \text{ kg}$	$m_p = 1.67 \times 10^{-27} \text{ kg}$	$e = 1.6 \times 10^{-19} \text{ C}$
$k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$	$\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$	$\mu_0 = 4\pi \times 10^{-7} \text{ T} \cdot \text{m/A}$
$\mu_B = 9.27 \times 10^{-24} \text{ J/T}$	$c = 3 \times 10^8 \text{ m/s}$	$g = 9.8 \text{ m/s}^2$
f = femto = 10^{-15}	p = pico = 10^{-12}	n = nano = 10^{-9}
μ = micro = 10^{-6}	M = Mega = 10^6	G = Giga = 10^9
Sphere: $A = 4\pi R^2$	$V = \frac{4\pi}{3} R^3$	

8. A uniform current flows through a straight wire with circular cross section of radius R . At what distance from the center of the wire is the magnitude of the magnetic field one third of the magnetic field at the surface of the wire?

(1) $R/3$ and $3R$ (2) $R/3$ and $\sqrt{3}R$ (3) only $3R$ (4) only $\sqrt{3}R$ (5) $R/\sqrt{3}$ and $\sqrt{3}R$

9. In the RL circuit shown, the component values are $R = 15\ \Omega$, $L = 30\text{ mH}$, and $\mathcal{E} = 9.0\text{ V}$. If the switch is closed at $t = 0$, what is the magnitude of the potential difference across the resistor at 3.0 ms ?

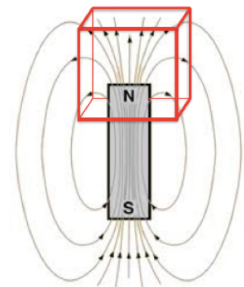


(1) 7.0 V (2) 9.0 V (3) 4.5 V (4) 3.3 V (5) 5.6 V

10. An ideal transformer provides an output emf whose magnitude is 9 V and frequency 60 Hz , when the amplitude of the input is 180 V . It is connected to a resistive heater with a resistance $R = 100\ \Omega$. What is the equivalent resistance seen by the EMF source on the input side?

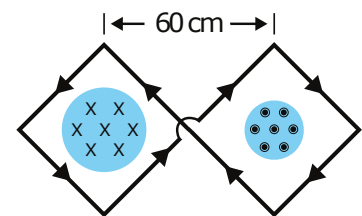
(1) $40\text{ k}\Omega$ (2) $100\ \Omega$ (3) $1\text{ k}\Omega$ (4) $2\text{ k}\Omega$ (5) $800\ \Omega$

11. The north pole of a magnet is enclosed by a Gaussian surface in the form of a cube. The magnetic flux is 0.5 Tm^2 through the top face of the cube and -2.5 Tm^2 through the bottom face. If the fluxes through each of the four side faces are identical to each other, what is the magnetic flux (in Tm^2) through one of those side faces?



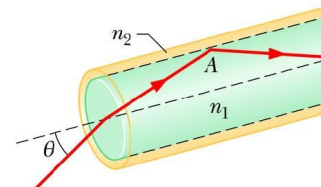
(1) 0.5
(2) 2.0
(3) -2.0
(4) 3
(5) -3

12. The figure shows two circular regions — region 1 of radius $r_1 = 25\text{ cm}$ and region 2 of radius $r_2 = 20\text{ cm}$ — separated by 60 cm . The electric field in region 1 is 500 N/C going into the plane of the page, and that in region 2 is 700 N/C coming out of the plane of the page. The magnitudes of both fields are decreasing at a rate of $50\text{ N/(C}\cdot\text{s)}$. Calculate the magnitude of the path integral $\oint \vec{B} \cdot d\vec{s}$ along the path drawn in the figure; ignore the bending of the path at the crossing.



(1) $1.8 \times 10^{-16}\text{ Tm}$ (2) $4.3 \times 10^{-12}\text{ Tm}$ (3) $8.2 \times 10^{-15}\text{ Tm}$ (4) $6.2 \times 10^{-19}\text{ Tm}$ (5) $7.8 \times 10^{-9}\text{ Tm}$

13. The figure depicts a simplistic optical fiber: a plastic core ($n_1 = 1.58$) is surrounded by a plastic sheath ($n_2 = 1.53$). A light ray is incident on one end of the fiber at angle θ . The ray is to undergo total internal reflection at point A, where it encounters the core-sheath boundary. (Thus there is no loss of light through that boundary.) What is the maximum value of θ that allows total internal reflection at A?



(1) 23.2° (2) 27.3° (3) 15.7° (4) 31.4° (5) 19.9°

20. Two stars that are close together are photographed through a telescope. The black and white film is equally sensitive to all colors. Which situation would result in the most clearly separated images of the stars?
- (1) Large lens, blue stars
 - (2) Small lens, red stars
 - (3) Small lens, blue stars
 - (4) Large lens, red stars
 - (5) Large lens, one star red and the other blue